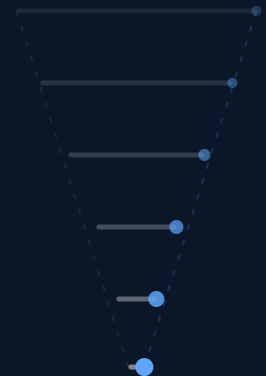


FINTECH PRODUCT ANALYTICS

Investment Journey

Event Instrumentation · Product Thinking · SQL

A product analytics case study focused on understanding, instrumenting, and measuring a fintech investment onboarding journey.



PRODUCT CONTEXT

Product & Platform Overview

A fintech investment platform helps users build long-term wealth through Systematic Investment Plans (SIPs). Users invest recurring amounts into diversified assets such as equities, gold, and fixed deposits. The platform aims to keep investments accessible while enabling users to build long-term wealth — allowing users to access liquidity against their investments without selling assets or breaking their SIP.

CASE STUDY OBJECTIVE

Analytical Focus

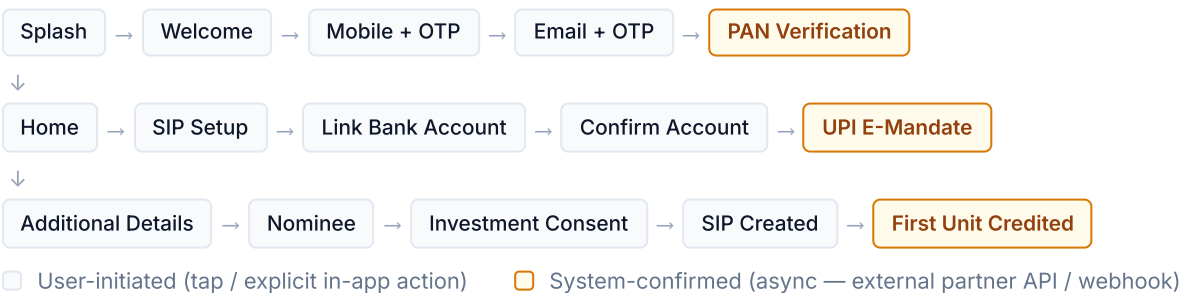
As the product analyst evaluating the investment onboarding journey, you are tasked with:

- **Event Instrumentation:** Instrument the journey while distinguishing user actions from system-confirmed outcomes.
- **Product Thinking:** Form a testable drop-off hypothesis and define North Star and counter-metrics.
- **SQL Validation:** Validate journey progression, completion time, and drop-offs using PostgreSQL against event data.

THE JOURNEY

Investment Onboarding Flow

The complete onboarding journey from app open to first investment unit credit is shown below. Amber highlights indicate externally confirmed outcomes where explicitly represented in the journey.



Asynchronous / System-Confirmed Stages: External confirmation is central to fintech onboarding. **PAN verification** is recorded when the KYC API confirms successful identity verification. **Bank account verification** (`bank_verified`) is recorded when the banking partner confirms successful verification (following the user's "Confirm Account" action). **UPI mandate approval** is recorded when the payment gateway or bank confirms mandate success. **First unit credit** is recorded when the Registrar and Transfer Agent (RTA) confirms unit allocation (which may occur hours or days post-creation). Analytics triggers must capture external confirmations rather than in-app button taps.

TASK 1

Event Specification

Design the analytics instrumentation for the investment journey, from app open to first successful unit credit. Every event must earn its place and every property must answer a real analytical question.

HARD CONSTRAINT — MAXIMUM 12 EVENTS: You will almost certainly identify more than 12 potential instrumentation points. Cut ruthlessly, prioritise events providing the highest analytical leverage, and justify your exclusions. Your selection strategy is a core part of the evaluation.

For each of your 12 selected events, provide:

Field	What is required
event_name	A consistent snake_case name applied uniformly across all 12 events.
Trigger	The exact moment the event fires, and whether it is user-initiated (direct tap/action) or system-confirmed (API response, webhook, external partner confirmation). Describing a user state is insufficient (e.g. <i>"User selects amount"</i>). Specify the precise firing point (e.g. <i>"fires on tap of Confirm SIP button"</i> or <i>"fires upon 200 OK webhook from PAN verification API"</i>).
Properties	The specific properties attached to this event, and why each earns its place. State what analytical question each property answers. Exclude properties that do not inform product decisions.
Question it answers	One concise sentence stating the exact product or business question this event answers. If you cannot name the question, drop the event.
If users drop here	What a drop-off at this step reveals about user behavior or product friction, and one specific product change to address it.

Excluded Events & Selection Strategy

After specifying your 12 events, provide:

- A table of events considered but excluded, with a clear, defensible reason for each exclusion under the 12-event constraint.
- A concise statement of your **selection strategy** — the overarching principle you applied to decide what to track and what to cut.

TASK 2

A Hypothesis You Could Be Wrong About

Target length: half a page, no more.

- Which single stage in the onboarding journey do you believe leaks the most users, and why?
- What evidence in the event data would prove your hypothesis wrong?

This section evaluates your ability to state a **specific, disprovable hypothesis** rather than hedging across possibilities. Name one stage, provide reasoning grounded in fintech failure modes or friction, and state the exact data condition that would falsify your position.

TASK 3

North Star Metric & Counter-Metric

Constraint: Five sentences total.

- **North Star Metric:** One primary metric for the investment journey representing genuine value delivered to the user, not merely intermediate step completion.
- **Counter-Metric:** One metric that would deteriorate if the team optimized the North Star poorly or gamed it, guarding an uncaptured dimension of user experience.

TASK 4

SQL Analysis

The synthetic events table below records every tracked action in the investment onboarding journey. Write PostgreSQL queries that you believe would run correctly against it. You will not execute the queries against a live database — clean syntax and logic are evaluated.

Schema

Column	Type
event_id	text
user_id	text
event_name	text
event_ts	timestamp
properties	jsonb

TASK 4

Data Profile

The table below profiles the synthetic events dataset available in the database. Total rows and distinct user counts are exact:

event_name	rows	distinct users
app_open	12,410	4,102
mobile_verified	2,980	2,974
pan_verified	2,341	2,338
sip_amount_selected	14,205	2,290
bank_verified	1,105	1,098
mandate_completed	742	731
additional_details_saved	690	688
investment_consent_verified	655	640
sip_created	631	631
first_unit_credited	588	588

Important: `mandate_completed` and `first_unit_credited` are recorded when external partners confirm success, not when the user acts inside the app.

Analysis Requirements

Provide the PostgreSQL analysis as a single labelled `.sql` file containing the following three outputs:

- Funnel progression:** Produce a step-by-step funnel showing distinct users reaching each stage and step-over-step conversion from the preceding stage.
- Median completion time:** Calculate the median elapsed time between the first and last stage for users who completed the full journey.
- Largest drop-off:** Identify and name the single consecutive stage transition with the largest absolute drop-off in distinct users.

Assumptions: The data and brief leave several aspects ambiguous. List every assumption made regarding repeated events, timestamp ordering, journey restarts, or completion criteria. This section carries significant weight in analytical rigor.

Case Study Ground Rule: AI tools are allowed. You may use ChatGPT, Claude, or other tools you normally work with. However, every analytical choice, assumption, event definition, SQL query, and conclusion should be understood and defensible by you. The use of AI does not replace analytical understanding.